An Automated Tagging System for Arbitrary Documents

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Problem Statement

This programming project seeks to challenge the adage that "you cannot judge a book by its cover" by designing a predictive model capable of applying the complex, multi-dimensional tagging system inspired by the Archive of Our Own website to arbitrary text documents. The advantage of such a tagging system over the conventional summary system that Large Language Models already excel at is the ability to apply binary filters to a massive sample of text for the purposes of identifying important people, sensitive content, and the like. This information can be stored as metadata to facilitate an efficient and human readable document filtering system.

Process

A Huggingface user by the pseudonym, nyuuzyou, posted a corpus of scraped Archive of Our Own works and their associated metadata, including the desired tag information to be used in this project. Early in project development, the possibility of utilizing a smaller set of data collected by automated scraping scripts for ease of use and a more manageable scale was considered. However, this method proved to be infeasible due to a combination of restrictions on Google Colab's resource limits, and from scraping countermeasures implemented by Archive of Our Own in response to mass scraping. The training program had to be run in Google Colab while mounted to my personal Drive account in order to access a copy of this data which I produced.

Design

The core design involves fine tuning on Meta's Bidirectional AutoRegressive Transformer (BART) Large Language Model architecture on this corpus, consisting of a sample of texts and metadata tags, to algorithmically model the transformation between prose and specific tags. This particular architecture was selected because it is an Encoder-Decoder model, which specializes in tasks related to summarization and translation, both of which are similar to the objective of this project. Additionally, BART is open-source, which is required for fine-tuning. Based on the model card, BART seems to be designed for correcting errors in text. Personal testing showed that the model produces 'corrected' versions of the input text in early stages of fine-tuning, which corroborates that idea.[1] If more time and resources were available, I believe it would be worthwhile to investigate whether another model might be better suited for this project.

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[1] Hugging Face. (n.d.). *BART*. Transformers documentation. Retrieved May 1, 2026, from https://huggingface.co/docs/transformers/en/model_doc/bart

Some of the tags contained within Archive of Our Own metadata are categorical, while others are strings chosen by the submitter. Transforming text into categorical labels is the simpler task, and is therefore well suited to encoder only architecture. The code repository was partitioned into separate notebooks containing separate programs for fine tuning classification on BART's encoder, and generation on BART's encoder-decoder architecture. The majority of the time spent on this project and report will refer to the generation program.

Fine-tuning was selected over other adaptation methods primarily as a matter of scope. Although zero-shot and prompt-engineering tests using both Claude and Gemini were both promising and effective, these were brief solutions that relied on the massive scale of these existing models to replicate a relatively simple task. Retrieval Augmented Generation introduced a layer of complexity and rigidity that I decided was unnecessary for what was ultimately a specialized type of summarization task. Hypothetically, if this model were to be used as an actual application, a small fine-tuned local model would be able to fine tune itself to the prose choice and labelling preferences of an individual user, so fine-tuning was ultimately selected as the adaptation method.

Evaluation

The baseline metric of success is the model's classification accuracy and the percentage of correctly predicted existing labels for Archive of Our Own sourced test data. One critical question that this report intends to answer is whether there is a relationship between the token-by-token loss, which the model trains to minimize, and the veracity or formatting of the statements generated, which there is no clear way to perform gradient descent on directly. There's no clear analogue for false positive or false negative rate for measuring the information generated by a Large Language Model, so accuracy will be defined as the proportion of labels which the model generates which are also in the set of true labels, as well as the proportion of labels in the set true labels.

During the training process, loss decreased by approximately a factor of two, indicating an improvement over the initial model. Curiously, training loss was consistently greater than validation loss by more than a factor of four, which is unusual and may indicate a problem with the training process itself.

Epoch	Training Loss	Validation Loss
1	No log	12.104989
2	57.616357	9.431197
3	40.188788	6.975745
4	31.511786	6.174611
5	26.656497	5.594237
6	26.656497	5.295691
7	24.664096	5.136044
8	23.606227	5.053482
9	23.139053	5.069856
10	22.948965	5.008242

However, loss is measured at the token level, and loss did not appear to correspond with an increase in labelling accuracy as defined above. When loss plateaus around the eight training epoch, the generated text was not comprehensible as English. Curiously, the phrase GoldMagikarp appeared in a few of the generated tags.

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rapes and bruises and a couple of burns, living with Papa definitely made that a possibility.

dincabledabled �abled detrimabledabledGoldMagikarpabledabled helicopabledabledabledabled disc

me off to book their weekend away. It wasn’t like they couldn’t afford it. The only thing whic
usceptabledabledabledabledincabledabled �abledabledGoldMagikarpabledabled helicopabledabledab
```

The fact that the generated text repeats similar tokens and phrases, but not deterministically, seems to indicate that the model is overfitting to patterns that are not useful to the task. This points to two major technical limitations. The first is the small amount of training data used, which makes it harder to distinguish patterns across every datapoint from random noise. The other is the LoRA rank, which restricts the maximum complexity of the model and therefore the final accuracy to the data. Both of these were restricted to prevent prohibitively long training time.

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--- Sample 5 ---
Original Text: lee minho knew better than anyone else: he was an idiot for let
True Tags:     Han Jisung | Han/Lee Minho | Lee Know
Generated Tags:  suscept suscept suscept suscept suscept suscept  suscept susc

--- Sample 6 ---
Original Text: Currently, Norman’s life was pretty fucking amazing. He didn’t
True Tags:     Ethan Mars/Norman Jayden
Generated Tags:  suscept suscept suscept suscept suscept suscept  suscept susc

--- Sample 7 ---
Original Text: (No ones pov)A normal day of school where dicks will be dicks,h
True Tags:     Denmark/Norway (Hetalia)
Generated Tags:  suscept suscept suscept suscept suscept suscept  suscept susc
```

In the pattern matching sense, this model has 0% informational accuracy. It failed to accurately identify themes, tone, or even major characters in novel text as anticipated.

Conclusion

At the scale of the data utilized, BART's attention mechanisms on LoRA fine-tuning were not sufficient to model the format or information contained in Archive of Our Own's tagging system. Despite the decrease in token prediction loss during training, the generative outputs are seemingly random noise regardless of the input text. Knowing that prompt engineering on industry leading Large Language Models is known to be effective at creating summaries in a desired format, I see no reason to believe it is impossible for a small locally run model to do the same given sufficient compute power and time.

Acknowledgement

Project code was partially generated using Gemini LLM

General Python knowledge was sourced from https://www.w3schools.com/python/default.asp

Base Model card: https://huggingface.co/facebook/bart-base

The data source appears to have become unavailable in the past few weeks:

https://datafish.ru/nyuuzyou/datasets/ao3/files